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Exp-15: Implement Iris Flower Classification using Naive Bayes classifier

AIM

To implement Iris flower classification using the Naive Bayes classifier and classify iris flowers into different species based on their sepal and petal measurements.

ALGORITHM

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Separate the dataset into input features X and target labels Y.
4. Split the dataset into training and testing data.
5. Create a Gaussian Naive Bayes classifier.
6. Train the classifier using the training data.
7. Predict the classes of the test data.
8. Compare the predicted values with the actual values.
9. Calculate the classification accuracy.
10. Display the confusion matrix and classification report.
11. Use the trained model to classify a new iris flower.

PROGRAM

```
# Import required libraries

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix

# Load the Iris dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split the dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create the Naive Bayes model

model = GaussianNB()

# Train the model

model.fit(X_train, y_train)

# Predict the test data
```

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y_pred = model.predict(X_test)

# Display results
print("Actual Labels:")
print(y_test)

print("\nPredicted Labels:")
print(y_pred)

# Accuracy
accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy: {:.2f}%".format(accuracy * 100))

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

```

Output:

```

... Actual Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Predicted Labels:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Accuracy: 100.00%

Confusion Matrix:
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]

```

SAMPLE INPUT

For a new iris flower:

Feature	Value
Sepal Length	5.1 cm
Sepal Width	3.5 cm
Petal Length	1.4 cm
Petal Width	0.2 cm

$$P(A | B) = 0.68$$

$$P(A)$$

$$P(A)$$

$$P(B | A)$$

$$P(B | A)$$

$$P(B | \neg A)$$

$$\frac{P(B \mid \neg A)}{P(B)}$$

$$\frac{A}{\neg A}$$

Give feedback

SAMPLE OUTPUT

Iris Flower Classification using Naive Bayes

Predicted Class: Iris-setosa

Actual Class: Iris-setosa

Accuracy: 96.67%

Confusion Matrix:

[[10 0 0]

[0 9 1]

[0 0 10]]

RESULT

The **Iris Flower Classification** was successfully implemented using the Naive Bayes classifier. The model successfully classified iris flowers into **Setosa, Versicolor, and Virginica** with high accuracy.

